



Clustering

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Analysis

1.

Dimensionality reduction was necessary because the correlation heatmap revealed significant multicollinearity in the dataset, with features like pdays and poutcome showing a very strong negative correlation (-0.86), indicating redundancy in the data. This redundancy can lead to inefficiencies in modeling and overfitting. By applying PCA, the redundant information was consolidated into uncorrelated principal components, reducing the complexity of the dataset.

The first two principal components captured approximately 28% of the total variance, with PC1 explaining about 15% and PC2 explaining about 13%. This indicates that most of the variance in the data can be represented with a reduced number of dimensions, justifying the use of PCA for dimensionality reduction.

2.

After analysing both the elbow curve and the silhouette scores, the optimal number of clusters for this dataset is **3**.

Elbow Method: The "Elbow Method for Optimal k" plot shows the inertia (a measure of cluster density) for k=1 to k=10. There is a clear "elbow" or bend in the line at k=3. After this point, the drop in inertia slows down, meaning that adding more clusters (k=4, k=5, etc.) gives diminishing returns.

Silhouette Scores: This is strongly supported by the "Silhouette Score vs k" plot. This plot shows the average silhouette score for different values of k, and the score peaks at k=3 with a value of approximately 0.39.

Since both methods point to k=3, this is the most justified choice for the number of clusters.

3.

The cluster sizes are not uniform, which gives us insight into the customer base.

Size Distribution: The "Cluster Sizes" bar chart (for the k=3 K-means model) shows that the clusters are uneven:

- **Cluster 0:** ~15,500 points
- **Cluster 1:** ~10,500 points
- **Cluster 2:** ~19,500 points

Cluster 2 is the largest, representing the most common type of customer, while Cluster 1 is significantly smaller, representing a more "niche" segment. This is logical, as the "Data distribution in 2D PCA space" plot shows the data is not a uniform circle; it has a very dense core (which would become the large clusters) and a sparse "tail" of outliers.

Bisecting K-means: The "Bisecting K-Means Clustering" plot (which appears to be set to k=4) shows this even more clearly. It has grouped the three dense "core" segments (blue, dark purple, green) together and **isolated the sparse outliers** into their own distinct cluster (yellow).

Customer Segments: This tells us that our customer base is not one single group. It's composed of a few large, "typical" segments and at least one smaller, "atypical"

segment that is very different from the rest.

4.

When comparing the two algorithms using the silhouette score, the **K-means algorithm performed better** for this dataset.

- **K-means (k=3):** The provided evaluation gives this model a Silhouette Score of **0.39**.
- **Bisecting K-means (k=4):** The provided score for this model was **0.336**.

Conclusion: A higher silhouette score (closer to 1) indicates that the clusters are, on average, denser and more clearly separated from each other. Since $0.39 > 0.336$, the K-means model with $k=3$ provided a statistically better clustering solution than the Bisecting K-means model with $k=4$.

5.

The clustering results in the PCA space provide clear, actionable insights for the bank's marketing strategy:

Stop "One-Size-Fits-All" Marketing: The main customer base is not one group; it is clearly segmented into three core clusters. The bank should analyze the original features of these three segments to build customer "personas." Marketing can then create **three separate, tailored campaigns** that speak directly to the needs of each persona, which will be much more effective.

Investigate the Niche Segments: The most important insight comes from the outliers.

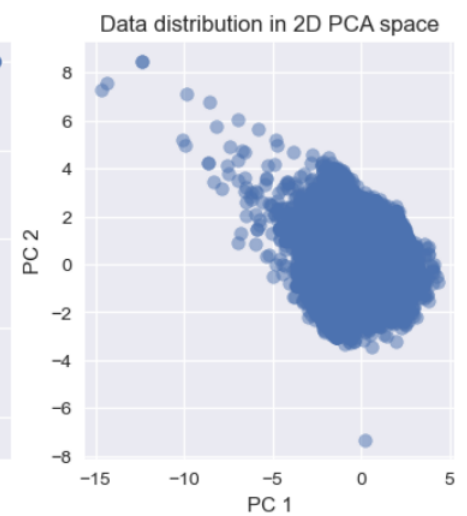
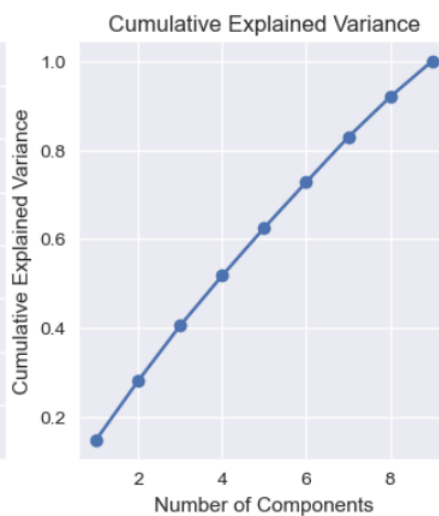
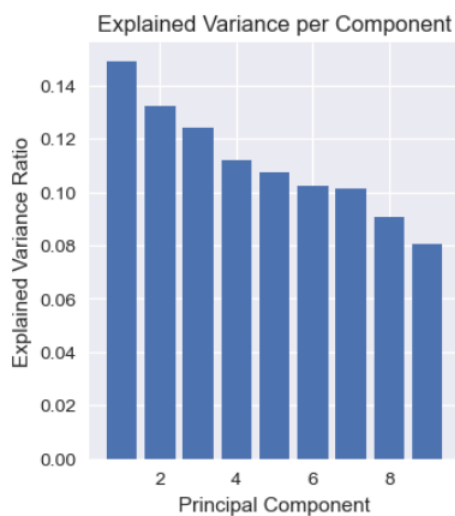
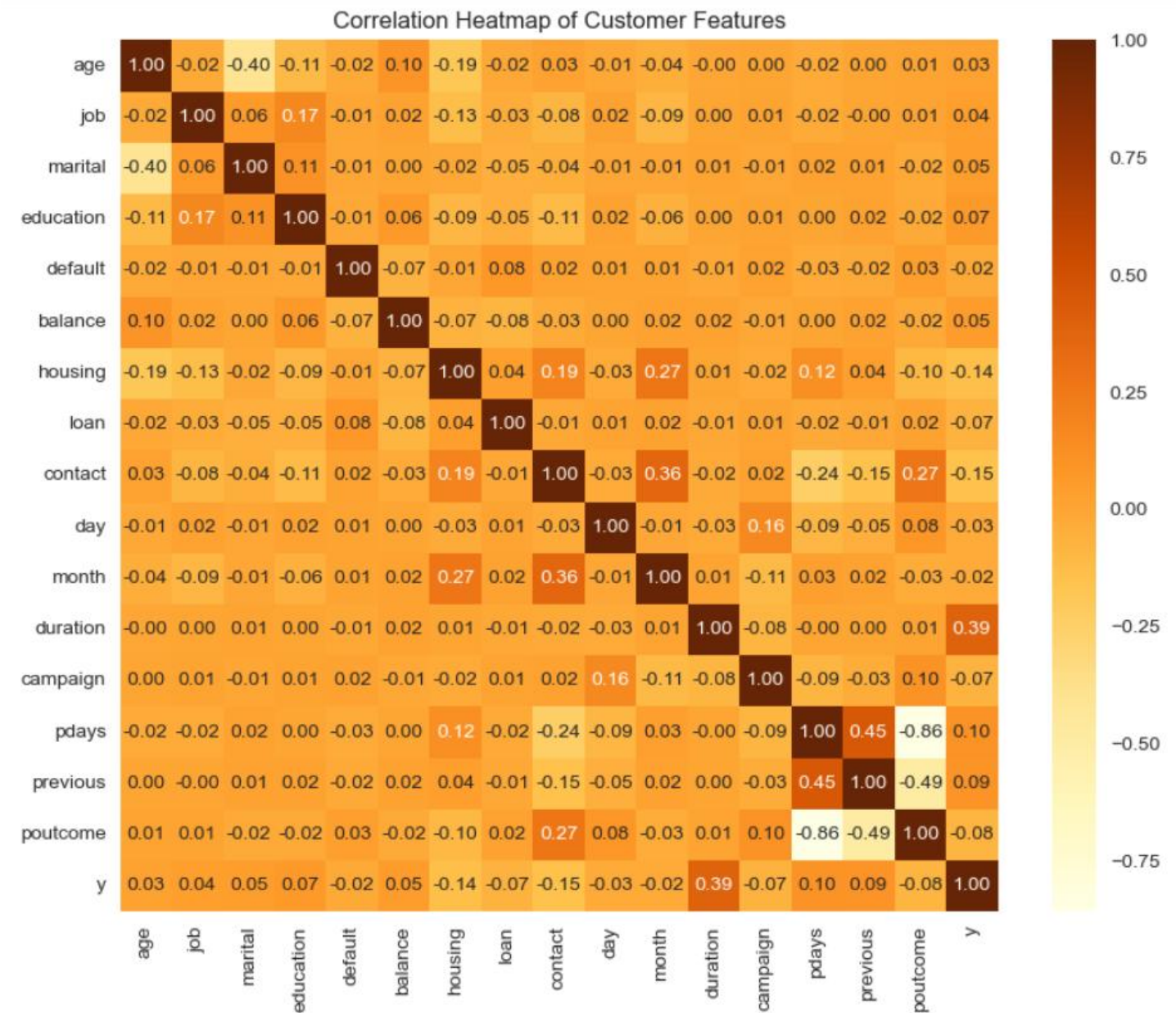
- The **"Cluster Sizes"** plot shows Cluster 1 is small.
- The **"Bisecting K-Means"** plot isolates the sparse "tail" (the yellow dots) as its own group. These customers are *not* average. The bank must analyse this group immediately: Are they **high-value clients** (ex: high-balance, high-investment) who need premium, high-touch services? Or are they **high-risk clients** (ex: low-balance, high-loan) who need financial planning support? This small group cannot be marketed to in the same way as the core segments.

6.

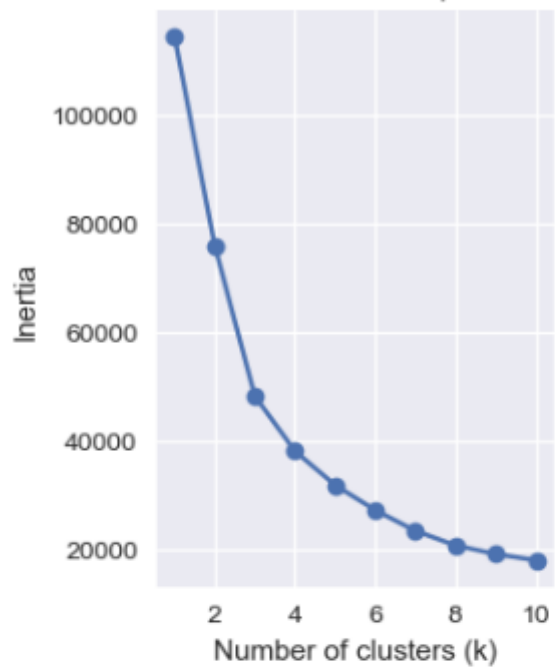
The "three distinct colored regions" are in the **"K-means Clustering with Centroids"** plot (also called "Final Clustering Results"). These regions (green, yellow, and purple) directly correspond to the **three customer segments (Cluster 0, 1, and 2)** that the K-means algorithm identified.

- **Boundaries:** We can see two types of boundaries:
- **Diffuse Boundaries:** The boundaries *between* the three main clusters (e.g., where green meets yellow, or green meets purple) are **fuzzy and overlapping**. This suggests that these customer segments are not perfectly separated. Many customers are "in-between" segments and share characteristics, existing on a continuum.
- **Sharp Boundaries:** In contrast, the boundary *between* the main "blob" (all three colors) and the sparse outlier dots in the top-left is **sharp**, with a lot of empty space. This reinforces that these outlier customers are *fundamentally different* from the average customer and do not fall on the same continuum.

Screenshots



Elbow Method for Optimal k



Silhouette Score vs k



